

Queens model assessment: expanded illustrated edition

Comparing GPT-5.6 Sol, Claude Opus 4.8, Grok 4.6, Gemini 3.1 Pro Preview and TypeSafe Jev 1.13. Expanded September 19, 2026 from the illustrated September 18, 19:21 MDT edition. **Solutions remain intentionally revealed**, as in that foundation report. No new contestant inference was performed for this editorial revision.

Executive summary

The four reasoning-model entrants received **one fresh API request per board**, containing the same blank image and instructions. They had no agentic sandbox, code execution, repository access, solver, browser, hints or repair turns. Eleven of twelve scheduled trials returned mathematically correct written solutions; nine also satisfied the strict JSON-only format. The missing twelfth answer was a Grok timeout, not a demonstrated wrong placement.

GPT-5.6 Sol had the strongest observed combination in this small pilot: 3/3 strict successes, a 13.70-second median outcome time and \$0.034088 total inference cost. Opus and Gemini also solved all three boards mathematically, each with one formatting failure. Grok solved two; its 7×7 answer was not captured before the approximately 601-second timeout. **Grok was reached through OpenRouter's Amazon Bedrock route**, following a rejected direct-xAI route.

Jev was evaluated separately because it accepts textual state and typed choices, not images or ordinary chat generation. Initial constructions and a broad candidate test did not solve the puzzles. Later, on the 5×5 board, Jev chose the correct solution in **6/6 calls when code exposed each candidate's occupied region letters together and asked for A through E exactly once**. All other constraints had already been enforced by the candidate generator. This demonstrates assisted recognition of the remaining condition, not unaided whole-puzzle planning.

The later experiments extend Jev's explicit-region candidate selection to **24/24 larger-board trials**, including six with 255 options. On six new boards, matrix and prose construction each scored **0/6 correct**, while Choice, Noul ranking and Score ranking selected the true region-list candidate **6/6** each. Ranking did not ensure correct absolute acceptance or precise ordinal scores.

All recorded inference charges now total **\$0.695604900**. Including unresolved allowances for routing errors, the Grok timeout and a later transport failure, the conservative ledger is **\$1.014599966**, leaving **\$1.985400034** of the original \$3 cap. These totals exclude host subscriptions, researcher time and local compute. No additional paid evaluation was performed to create this report.

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Edition lineage and what changed

This is a direct revision of the user-specified [illustrated foundation PDF](#), not a replacement by the later Jev-only narrative. The [foundation Markdown snapshot](#) and original PDF are preserved. All five original explanatory panels, board solutions, twelve individual reasoning-model outcomes, routing details and initial Jev experiments remain in this edition.

The separate [Jev-focused synthesis](#) supplied the larger-board scale, six new boards, primitive comparisons and revised ledger. Earlier statements that no larger-board candidate tests had occurred are superseded by the dated later evidence, not erased from the preserved foundation edition. Three GPT-5.6 Luna Xhigh research agents reviewed online sources for model identities, Jev interfaces, and Queens rules/play destinations before this PDF was regenerated.

How Queens works

This is **regional Queens**, not the ordinary chess N-queens problem. Each square has a row, a column and a colored region. Matching letters mean the same region. On an $N \times N$ board, place N queens so that:

1. Every row contains exactly one queen.
2. Every column contains exactly one queen.
3. Every colored/lettered region contains exactly one queen.
4. No two queens touch, even at a corner. Long diagonals are allowed when queens are more than one square apart.

See [LinkedIn's official Queens rules](#) and [play Queens online](#). Guest access and an independent practice alternative are described in the [online source guide](#); neither supplies our synthetic benchmark boards.

The important reasoning burden is global consistency. A queen can be locally legal now but leave no way to finish the remaining rows. Similarly, five individually plausible row choices can duplicate a column or region when assembled into one board.

Queens · 5×5 · Easy

Canonical region map, faithfully redrawn with letter labels and 1-based coordinates

Starting board

	1	2	3	4	5
1	A	A	B	B	C
2	A	B	B	D	D
3	A	E	B	D	D
4	A	E	D	D	D
5	E	E	D	D	D

Left: original blank puzzle structure. Right: the single placement confirmed by both solver paths.

Unique solution

	1	2	3	4	5
1	A	A	B	B	Q
2	A	B	Q	D	D
3	Q	E	B	D	D
4	A	E	D	Q	D
5	E	Q	D	D	D

Q queen

Figure 1. The original 5×5 target, faithfully redrawn for explanation, with its verified solution. A queen marker denotes a selected cell; letters and colors identify regions. The submitted blank PNG is retained separately as evidence, not replaced by this redraw.

The solution is **columns by row** [5, 3, 1, 4, 2], meaning row 1 column 5, row 2 column 3, and so on. Its occupied regions are C, B, A, D and E: each appears once. Consecutive queen columns differ by more than one, so no diagonal neighbors touch. This is the kind of written array the reasoning models were asked to return, not a picture of the answer.

This synthetic target was labeled “easy” because the fixed row-order reference search visited 38 partial placements. That is a reproducible property of one search ordering, not a validated rating of human difficulty. Exhaustive checking considered all 120 column permutations and found exactly one legal solution. [Original board and input audit](#).

The 7 by 7 target

Queens · 7×7 · Medium

Canonical region map, faithfully redrawn with letter labels and 1-based coordinates

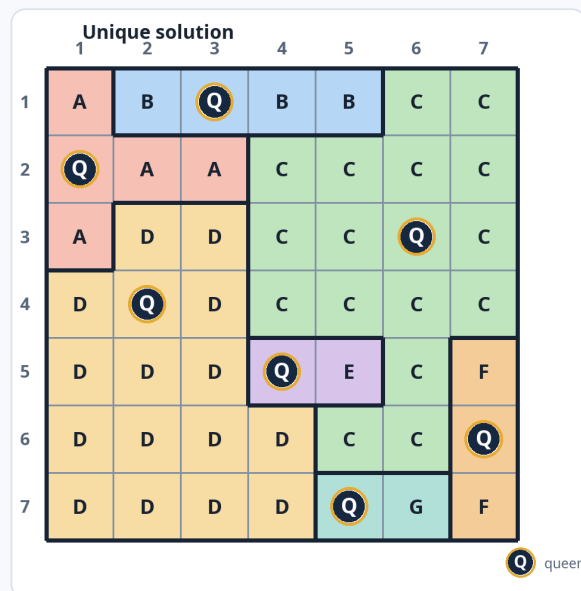
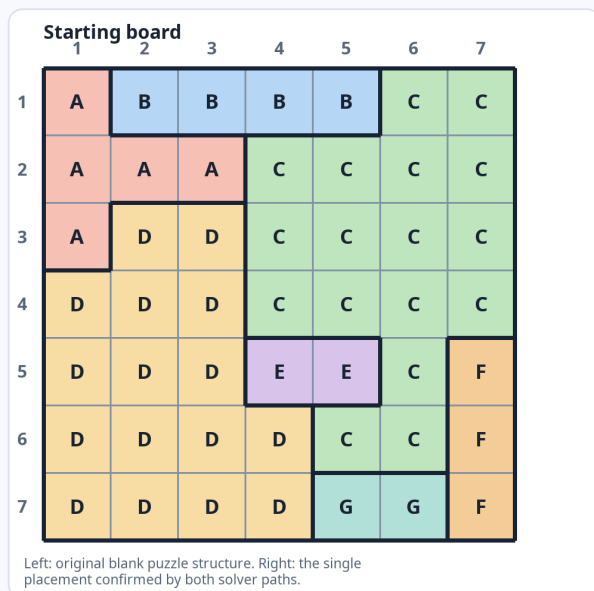


Figure 2. The same seven-region board used by every reasoning-model entrant, shown blank and solved. Region boundaries, cell membership and orientation are preserved in the redraw. The solved panel is reference material added for this report; it was not sent to the original competitors.

Seven queens must satisfy all four rules simultaneously. Looking only at rows is not enough: a choice can consume the sole useful column or region needed by another row. The solution shown was recovered from the repository's constraint solver and checked independently against the canonical region map; coordinates and hashes are retained in the [board verification record](#).

This board's provisional "medium" label corresponds to **197 row-order search nodes**, versus 162 under an alternative minimum-remaining-values ordering. Exhaustive checking covered all **5,040** column permutations and found one solution. Different algorithms doing different amounts of work on the same board illustrate why a difficulty label is not an intrinsic latency prediction.

What happened on this board

- Sol returned a valid, strict-format answer in 13.70 seconds.
- Opus returned a mathematically correct placement in 48.23 seconds, but extra formatting violated the exact JSON-only contract.
- Gemini returned a valid, strict-format answer in 86.63 seconds.
- Grok through Bedrock reached the client timeout after 600.964 seconds. No final answer or usage record was captured. It was not retried.
- Jev's native row-choice construction was interpretable but violated the puzzle constraints. It was a text-state experiment, not the identical-image condition above.

Reference solution: [3, 1, 6, 2, 4, 7, 5]. [Saved trial results](#).

The 9 by 9 target

Queens · 9×9 · Hard

Canonical region map, faithfully redrawn with letter labels and 1-based coordinates

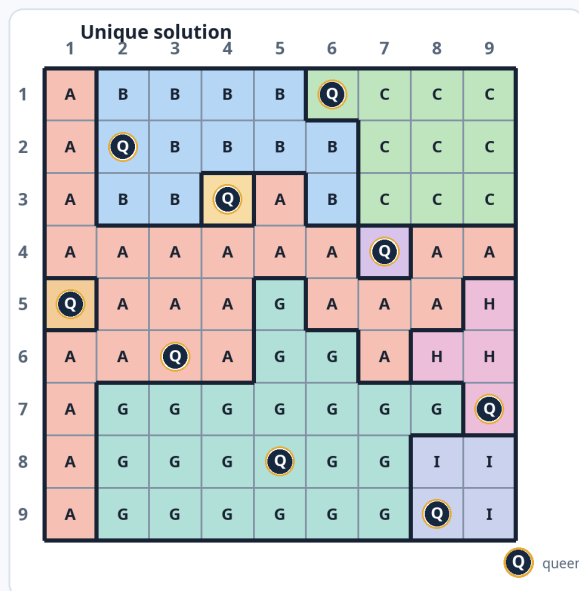
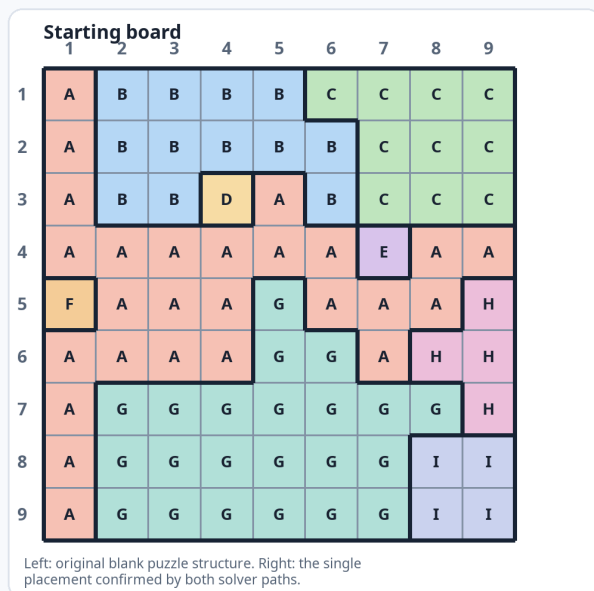


Figure 3. The nine-region target and verified solution. The single-cell regions are visible and materially affect the puzzle: their cells must contain queens. These are genuine features of the tested synthetic board, not extra clues introduced for this report.

The provisional “hard” label comes from **1,172 row-order search nodes**, versus 1,026 under the alternative ordering. Exhaustive checking considered **362,880** column permutations and confirmed one solution: [6, 2, 4, 7, 1, 3, 9, 5, 8]. The generator deliberately froze two single-cell seed regions; the final board actually has three singletons, D, E and F. This construction bias predates model calls and limits the difficulty interpretation.

What happened on this board

Sol, Opus and Grok returned strict-format correct solutions in 20.13, 39.13 and 52.82 seconds respectively. Gemini returned a mathematically correct answer in 71.97 seconds, but did not satisfy the JSON-only output contract. Jev's native row-wise text-state construction was invalid.

The larger board did not uniformly take longer than the smaller one. Grok's 9×9 response arrived while its 7×7 trial had timed out; Gemini took less time on 9×9 than 7×7. Board structure, model behavior, response length, routing and timing all matter. Three differently structured boards cannot establish a reliable size-to-runtime curve.

The three blank inputs were clean **768×768 synthetic PNGs with region letters**. They were not photos, live-game screenshots, or unlabelled color fields. Each reasoning model received the exact same PNG bytes for a given board, but provider image preprocessing and tokenization can still differ. [Frozen input protocol](#).

What each model actually received

The original language-model condition was one request, one fresh context and one scored answer opportunity per model per board. It was not an agentic sandbox. The API model named GPT-5.6 Sol was the contestant; the user's repository-connected Codex session was the evaluator, not a competing solver.

Condition	Input	Model work	External help during answering
Four reasoning models	Same image and rules per board	Generate the complete written placement	None; no tools or repair turns
Jev native row choices	Canonical text region grid and rules	Select one column for each row	Fixed typed options; simultaneous questions answered independently
Jev sequential 5×5	Text grid plus prior chosen queens	Choose the next row's column in five calls	Evaluator passes previous decisions; no backtracking
Jev candidate recognition	Text board plus supplied complete placements	Choose one option	Candidate construction and, in some formats, constraint features
Repository solver reference	Exact structured region map	Solve explicit Boolean constraints	Local Z3 solver; not a timed API contestant

The original 12 scheduled trials used `reasoning.effort: high`, `reasoning.exclude: true`, a shared `max_tokens: 12288` limit and non-streaming responses. No temperature override was set. Each user message contained the common written rules followed by the same base64 PNG for that board. There was no system history, solved example, code library or hidden answer key in the requests.

“One shot” does **not** mean one internal reasoning step. The reasoning models could use internal deliberation within that single call; the returned usage records report reasoning tokens. Excluding reasoning suppresses its returned content, not its computational cost. Effort names and a common token ceiling do not allocate identical compute across providers. [OpenRouter reasoning controls](#).

After two formatting failures, a deterministic post-hoc diagnostic extracted exactly one schema-shaped JSON object from the visible answer and checked the same rules. It did not call a model again, repair coordinates, choose among alternative answers or execute returned code. Thus **strict-format score** and **mathematical validity** answer different questions.

No separate autonomous GPT/Codex sandbox competitor, tool-enabled reasoning-model arm, or retry-with-feedback arm was actually run. Those remain proposed study conditions in the [comparative specification](#). The Luna agents preparing this report likewise are researchers, not entrants.

Reasoning model results

These figures describe three trials per model, not population-level accuracy. “Median outcome” includes the timeout duration for Grok. Cost totals use confirmed returned usage; Grok’s uncertain charge is separate.

Model	Strict JSON and correct	Written placement correct	Median outcome seconds	Confirmed USD
GPT-5.6 Sol	3/3	3/3	13.70	0.0340880
Claude Opus 4.8	2/3	3/3	39.13	0.2443050
Grok 4.6 via Bedrock	2/3	2/3	91.96	0.0755722
Gemini 3.1 Pro Preview	2/3	3/3	71.97	0.3130440

Individual outcomes

“Format only” means a unique visible JSON solution was correct under the post-hoc diagnostic but the full response broke the frozen output contract. “No answer” is not an invalid queen placement.

Model	Board	Outcome	Elapsed seconds	Confirmed USD
Sol	5×5	Strict pass	13.35	0.0089220
Sol	7×7	Strict pass	13.70	0.0105880
Sol	9×9	Strict pass	20.13	0.0145780
Opus	5×5	Strict pass	26.89	0.0567350
Opus	7×7	Format only	48.23	0.0976350
Opus	9×9	Strict pass	39.13	0.0899350
Grok via Bedrock	5×5	Strict pass	91.96	0.0333872
Grok via Bedrock	7×7	No answer at timeout	600.96	Unknown
Grok via Bedrock	9×9	Strict pass	52.82	0.0421850
Gemini	5×5	Strict pass	26.61	0.0412600
Gemini	7×7	Strict pass	86.63	0.1446640
Gemini	9×9	Format only	71.97	0.1271200

Sol was the least expensive and fastest by median outcome **in this run**. All eleven captured final responses contained valid placements, so among returned answers this pilot principally exposed formatting, latency and delivery differences. It does not establish a general intelligence ranking. [Primary grades](#), [separate format diagnostic](#).

Routes costs and operational failures

Model developer, inference provider and API gateway are distinct. **Every external evaluation request used OpenRouter through the local credential broker.** The provider route identifies the service selected behind OpenRouter; no direct Bedrock credentials were supplied to the evaluation client.

Exact requested model	Approved serving route	Input USD per million	Output USD per million
openai/gpt-5.6-sol	OpenAI	2.00	10.00
anthropic/claude-opus-4.8	Anthropic	5.00	25.00
x-ai/grok-4.6	Amazon Bedrock US West 2	2.20	6.60
google/gemini-3.1-pro-preview	Google AI Studio	2.00	12.00
typesafe/jev-1.13	TypeSafe	0.042	0.00

These are rates captured for the experiment, not a claim about current future pricing. Provider fallbacks were disabled. Grok's exact approved route tag was amazon-bedrock/us-west-2. Direct xAI had been planned at \$2/\$6 per million, but an account allowlist rejected that attempt before inference. The user explicitly approved Bedrock instead; prompt, image, model ID and generation settings stayed fixed. [Bedrock amendment](#), [provider routing documentation](#).

The 7×7 Grok timeout stopped execution overnight. No captured generation ID was available to recover its eventual upstream outcome. Continuation preserved the timeout and sent only the previously unstarted 9×9 trial the following day. It did not retry the failed board or give Grok another model's solution. Deferred ordering and different execution times limit latency attribution to the model itself.

Full inference ledger

Component	Confirmed USD	Unresolved allowance USD
Original reasoning-model pilot	0.667009200	0.225228800
First Jev deep dive	0.000792540	0.041268820
Jev 39-call 5×5 variations	0.002274384	0.000000000
Jev larger-board scale	0.020552112	0.040000000
Jev new-board primitive study	0.004976664	0.012497446
Total	0.695604900	0.318995066

The combined conservative ledger is **\$1.014599966**; **\$1.985400034 remains** under \$3. The original allowance contains \$0.107728 for the unmetered direct-xAI rejection and \$0.1175008 for the Grok timeout. The initial Jev allowance covers its unchanged-input rejection and two routing rejections. The later \$0.04 reserve covers the two original larger-board image-prompt rejections; \$0.012497446 covers one unmetered HTTP 520. These are reserved amounts, **not measured charges**. Final all-call billing remains unreconciled.

Jev's first 50 successful Decisions calls cost \$0.003066924; across all four Jev stages the total is now **136 answered calls, 691 typed decisions and \$0.028595700**, approximately **2.860 cents**. Six preserved compatibility/infrastructure failures bring all Jev attempts to 142. These are workload counts, not independent board sample sizes. Its output tokens were recorded but charged at zero. Host Codex/Luna subscription usage, local solver time, report preparation and any credit-purchase fees are outside this inference ledger.

Jev before candidate engineering

The user-provided [OpenRouter Jev Latest alias](#) resolved to [Jev 1.13](#) in the saved experiment. The alias is a moving pointer, not a permanent model version. Successful native responses consistently returned `typesafe/jev-1.13-20260917`, provider TypeSafe. The unchanged original image prompt was first sent as requested and received an HTTP 400: this family requires the separate Decisions API. That is an interface mismatch, not a wrong puzzle answer.

Jev accepts text/structured textual state and returns typed decisions such as Choice, yes/no Noul, or ordered Score. It does not generate an ordinary explanatory chat reply. The evaluator therefore used `/api/alpha/decisions`, fixed answer options and a text region grid. TypeSafe describes independent questions against shared state; selecting each row independently need not produce a globally consistent board. [System One](#), [Choice](#).

Native Jev test	Complete answer	API elapsed seconds	Confirmed USD
5×5 simultaneous row choices	Invalid	0.435	0.000056742
7×7 simultaneous row choices	Invalid	0.249	0.000088452
9×9 simultaneous row choices	Invalid	0.349	0.000129276
5×5 sequential rows, five calls	Invalid	1.189 total	0.000158802
5×5 selection from all 120 permutations	Invalid	0.245	0.000235746

The five rows are configurations, not five independent puzzles. Sequential state removed duplicate columns but still failed the region/non-touching requirements. The 120-permutation recognition choice satisfied row, column and non-touching rules, yet repeated regions. It was a meaningful spatial near-miss, not a formatting failure.

Other diagnostics: Jev read **9/9 sampled grid cells correctly**, but agreed with the checker on only **8/12** questions about its own candidate constraints. Those questions were unbalanced; an always-false baseline would score 9/12. They do not establish reliable self-verification. All eleven successful initial native calls were interpretable; their total cost includes these diagnostics. [Initial Jev audit](#).

Why the Jev placement fails

Jev before candidate engineering · canonical 5×5 target

Actual recorded 120-permutation recognition choice compared with the independently verified answer

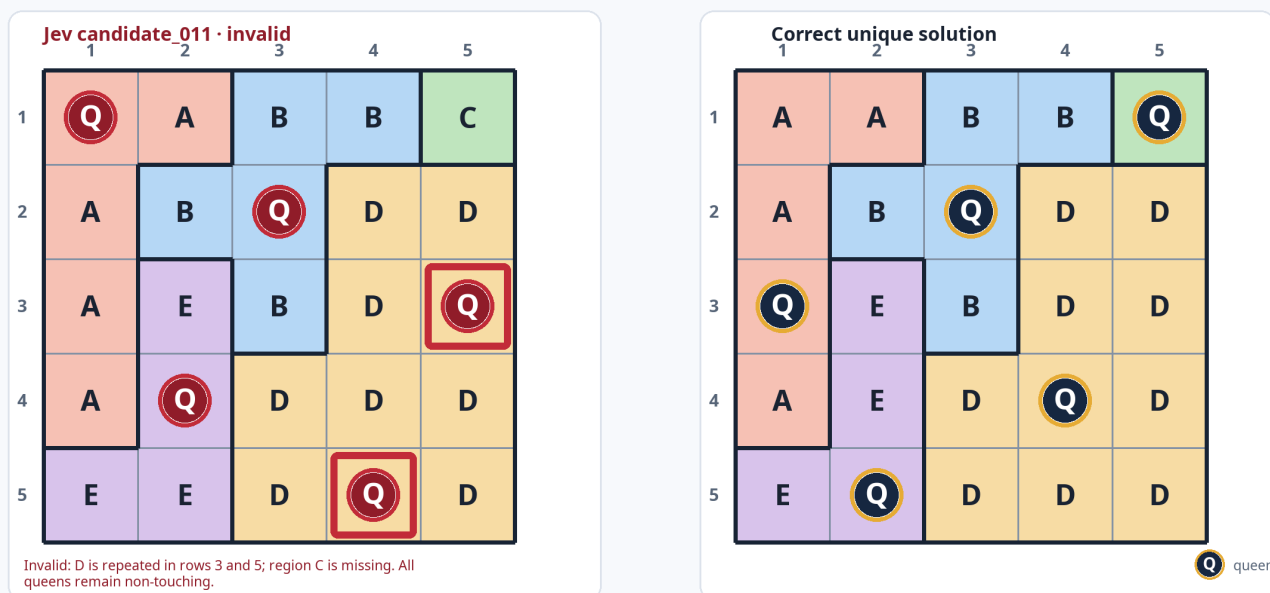


Figure 4. Jev's actual choice from all 120 column permutations in the initial deep dive, versus the unique legal solution. Red outlines identify the repeated-region queen cells. No queens touch in this particular wrong answer; region coverage alone makes it invalid.

Property	Jev selected placement	Unique solution
Columns by row	[1, 3, 5, 2, 4]	[5, 3, 1, 4, 2]
Occupied regions by row	A, B, D, E, D	C, B, A, D, E
One queen per row and column	Yes	Yes
Non-touching queens	Yes	Yes
Every region used once	No; D twice and C absent	Yes

The singleton region C in the upper-right cell must contain a queen. Jev's choice leaves it empty while putting two queens in D. Even though the array is well formed and the queens are well spaced, this one failed condition makes the whole answer incorrect. There is no partial credit on the competition's correct-solution objective.

This example also explains the later successful adaptation. If code supplies a candidate's region sequence explicitly, choosing between a repeated-letter sequence and a complete A-through-E set is a narrower problem than constructing the placement from the original board. The engine, not confidence or an LLM's opinion, establishes which array satisfies all rules. [Saved interpreted selection.](#)

Jev with engineered candidates

The later experiment supplied twelve candidates per question on the 5×5 target. A local oracle ensured exactly one complete solution was present but unmarked. Hard sets already obeyed row, column and non-touching rules; only region coverage remained. This is assisted recognition.

Input variation	Correct selections	Median API seconds
Column vectors with mixed invalid distractors	0/3	0.311
Column vectors with non-touching distractors	0/3	0.294
Individual queen coordinates and region labels	1/3	0.324
ASCII queen-board layouts	1/3	0.407
Precomputed region/column counts and touching pairs	3/3	0.314
Column vectors plus a solved different board	0/3	0.300
One given queen, globally completable next cell	0/3	0.267
Same, with candidate region labels	0/3	0.299
One given queen, any locally legal next cell	3/3	0.310
Flat list of occupied regions, explicit distinctness task	3/3	0.333
Same occupied regions in one sentence	3/3	0.233
Reordered fixed set, individual queen descriptions	1/3	0.248
Reordered fixed set, precomputed counts	3/3	0.310

All 39 requests returned interpretable Choice answers, with **0.3006 seconds median elapsed** and **\$0.002274384 total cost**. The first 27 calls were planned together. The final twelve were designed after their results; the successful flat-region reformulation is exploratory evidence, not a preregistered independent validation. [Full variation audit](#).

Why the successful prompt matters

The true target solution occupies **C, B, A, D, E**. The six successful flat-region trials asked for A through E exactly once, supplying the letters but no counts or validity flag. Coordinate lookup and all other rule filtering were performed by code.

In the count-rich trials, code computed every relevant check. Both original and reordered sets scored 3/3. Code could select the valid summary directly; Jev must not receive credit for that preceding search and validation.

Representation **and instruction specificity changed together**; formatting alone cannot be credited for the improvement. Overlapping candidate sets on one board do not establish general accuracy or calibration. No larger-board candidate tests had occurred at the foundation edition's cutoff; the later studies below now extend that evidence.

Partial moves and the solved example

A move can pass every immediate test yet make eventual completion impossible. That distinction explains why local next-cell selection and puzzle planning must be scored separately.

Three Jev tests fixed a known-correct target queen and offered twelve additional cells. Each set contained exactly one cell belonging to a complete solution, but respectively nine, five and five cells were locally non-conflicting. Jev selected locally legal cells in all nine partial-move calls, including the six explicitly requesting full completion. **None of those selected cells permitted a complete solution.**

Thus 3/3 local-task success and 0/6 global-task success are compatible. Local uniform random choice would average 52.8% accuracy in those sets. The supplied true queen and carefully constructed options are substantial scaffolding; this is not a natural unrestricted next-move test.

Few-shot context example · 5×5

Different connected-region board from the Jev variation study; not a scored target

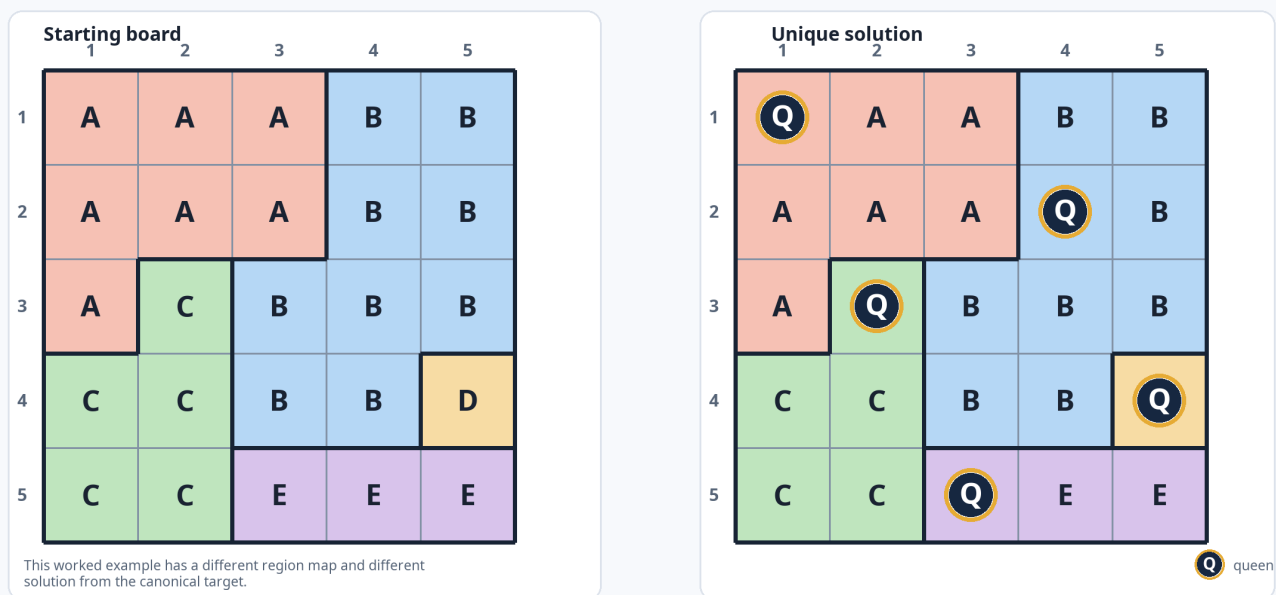


Figure 5. The different 5×5 board used for few-shot context, with its unique solution. This is **not another scored target**. Its region map and solution differ from the target. In the actual Jev request the example was supplied as text with its solution and constraint features, not as this image.

The example and no-example requests otherwise had the same target options and wording. Both scored **0/3**. Mean probability on the correct target option fell from 0.08 to a provider-rounded 0.00, but this is one example and three closely related tests. It does not establish that few-shot examples generally harm Jev.

Fresh requests had no conversation history. The researcher tried different prompts across trials; Jev did not learn from a previous failed answer or receive corrective feedback. “Fresh instance” describes client context isolation, not independently initialized model weights, guaranteed backend process isolation, or verified absence of caching.

Jev candidate scale on larger boards

The scale study then used the original 7x7 and 9x9 boards. Every full candidate had one queen per row and column and no touching queens. Exactly one also covered every region once. Columns-only and explicit-region variants were paired on identical candidate contents and order.

Board	Options	Random-pick reference	Columns-only	Explicit regions	Mean probability on true option, regions
7x7	12	8.33%	0/3	3/3	0.960
7x7	50	2.00%	0/3	3/3	0.877
7x7	100	1.00%	0/3	3/3	0.837
7x7	255	0.392%	0/3	3/3	0.730
9x9	12	8.33%	1/3	3/3	0.900
9x9	50	2.00%	0/3	3/3	0.790
9x9	100	1.00%	0/3	3/3	0.730
9x9	255	0.392%	0/3	3/3	0.667

At 255 options, the count controls also scored 3/3 for each size. The maximum-size explicit-region requests used 19,093 input tokens for 7x7 and 22,289 for 9x9. Their median request times were 0.380s and 0.671s respectively.

No explicit-region selection failure was found before reaching the Choice interface limit. However, mean probability on the correct answer declined as options increased, and one successful 9x9/255 response assigned the true option only 0.50. A broader distribution is not itself a failure, but it prevents a blanket claim that scale had no effect.

The near-miss pools were finite. For 7x7 there were only 71 invalid candidates covering six regions; the 255-option set added 183 covering five. For 9x9 there were 114 invalid candidates covering eight regions; the largest set added 140 covering seven. Consequently, option count, input length, ordering and distractor mix were not perfectly isolated. [Scale assessment](#).

Jev primitives and text input

Jev 1.13 accepts text, not images, audio or video. Its state may be a string, JSON object or array. An image must be transcribed into text or structured data before Jev receives it. The tested OpenRouter route is the Decisions endpoint, not a generative chat interface. The gateway path in our retained requests was `/api/alpha/decisions`; TypeSafe's [direct API reference](#) documents its own `/v1/systemone` endpoint, not that OpenRouter path. [TypeSafe model specification](#).

Primitive	Question	Returned value	Interpretation
Choice	Which supplied option?	Selected option, option probabilities, confidence	Relative selection within a finite set; up to 255 alternatives
Noul	Is this statement true?	A number from 0 to 1	Model probability of yes, not a hard Boolean; no separate confidence field
Score	Where on defined ordered levels?	Expected level, level probabilities, legend, confidence	A position on the caller's rubric, not inherently a probability of validity

Choice and Noul differ in more than the number of alternatives. Noul asks for an absolute judgment of a statement; Choice asks for a relative winner among alternatives. A series of Noul judgments does not automatically enforce that exactly one candidate is valid. Choice can force a winner even if none is correct unless a rejection option is provided. [Choice documentation](#), [Noul documentation](#).

Score uses an ordered array of descriptive levels, currently two to ten. Its score is the probability-weighted mean of the level indices. In a three-level rubric, a score of 1 could mean all probability on the middle level, or equal probability on the lowest and highest levels. Those imply very different validity beliefs. Inspect the level distribution rather than interpreting `score/2` as probability of a valid solution. [Score documentation](#).

The new coverage rubric uses level 0 for at least two missing regions, level 1 for exactly one missing region, and level 2 for complete coverage. Since candidate construction guarantees all other rules, probability on level 2 is relevant to validity. The expected level measures something else: ordinal coverage quality under this chosen rubric.

Questions in a batch see the same state but are evaluated separately. That API behavior does not establish statistical independence between answers. Similarly, confidence describes concentration of a returned distribution; it is not proof the candidate satisfies the puzzle. [Primitives and batching](#), [Confidence](#).

Version and context-limit distinctions

OpenRouter's public listing uses `typesafe/jev-1.13`, while saved responses identify the tested build as `typesafe/jev-1.13-20260917`. TypeSafe's current model documentation names `jev-1.13.0`. These are different naming layers; no separate public page for the dated response build was verified. Preserve the request and response identities rather than rewriting them to a current alias. [OpenRouter listing](#), [TypeSafe models and aliases](#).

OpenRouter displays a 32K context limit. TypeSafe documents a 64K per-request budget with a tighter 32K limit on state plus the longest question. Do not treat those displays as one proven gateway limit or infer that the direct API's larger aggregate budget was available on the tested route. This evaluation reports actual saved token counts and observed compatibility, not a new limit test.

Independence and new held-out boards

The earlier lack of independent boards was a **limitation of the exploratory design**, not a reason to discard its paired prompt comparisons. It would become a statistical flaw if repeated representations of the same board were counted as independent puzzles or used to claim general reliability. The reports warned against that interpretation; the new study addresses some of the missing generalization evidence.

There are several different meanings of independence:

Dimension	What it requires	What this work establishes
Fresh context	No previous answer or conversation carried into the request	Used except in the explicitly sequential experiment
New puzzle instance	A genuinely different region partition and solution problem	Six new boards, excluding rotation/reflection and relabeling duplicates
Holdout discipline	Prompts and thresholds fixed before judging the new data	New corpus and all task definitions frozen before inference
Statistical independence	An appropriate sampling model and units without hidden dependence	Not guaranteed by API calls; analyze at board level and retain paired dependence
Distribution diversity	Puzzles representative of the intended population	Not established: the same synthetic generator family is reused
Backend isolation	Separate serving process or absence of caching	Not observable or guaranteed through this interface

Six new boards were generated with separate fixed seeds: two each at 5x5, 7x7 and 9x9. The first board meeting connected regions, unique solution and nonduplication was accepted for each seed. Acceptance did not depend on Jev performance. Exhaustive checks confirmed each solution; eight geometric symmetries plus region relabeling were considered when rejecting duplicates.

The new boards have singleton-region counts of 0 and 1 at 5x5, 1 and 1 at 7x7, and 2 and 2 at 9x9. The 9x9 generator intentionally restricts two seed regions from growing. This shared construction bias means the six boards are not an adequate sample of every kind of Queens layout. Size also must not be equated with a validated difficulty rating.

Each board receives five tasks: matrix construction, prose construction, columns-only candidate comparison, region-list candidate comparison, and an all-invalid region-list set. Candidate conditions reuse the same board intentionally so methods can be compared on matched evidence. These create **six board-level evaluation units**, not 30 independent tasks or 534 independent decisions.

Additional fresh requests on the old boards would mainly measure repeatability or ordering sensitivity. Additional **new, held-out boards from multiple generator or source families** are the useful next step for estimating generalization. After inspecting this study, its boards are no longer an untouched holdout for later prompt tuning.

Choice Noul and Score on new boards

The new study completed 30 answered requests: 12 construction calls and 18 mixed-primitive calls. One additional attempt returned HTTP 520 without an answer; a manually reviewed, identically prompted transport retry succeeded. No incorrect puzzle answer was retried. The raw error and budget reserve remain preserved.

Each mixed-primitive request has 12 full placements plus a Choice none option, 12 Noul validity questions, and 12 Score coverage questions. All candidates already satisfy row, column and non-touching rules. The region-list format supplies region lookups but no counts or validity labels. The all-invalid condition replaces only the true candidate; Jev is told that zero or one supplied candidate may be valid.

Construction and ranking

Task on six new boards	Correct
Matrix-text construction	0/6
Prose-text construction	0/6
Columns-only: Choice selects true candidate	0/6
Columns-only: true candidate uniquely highest Noul	0/6
Columns-only: true candidate uniquely highest Score	0/6
Explicit regions: Choice selects true candidate	6/6
Explicit regions: true candidate uniquely highest Noul	6/6
Explicit regions: true candidate uniquely highest Score	6/6

The positive recognition result therefore extends beyond the original three target boards. It remains a six-board result under a prepared-input protocol, not evidence of near-perfect performance on arbitrary Queens puzzles. Rankings required a unique maximum; ties were not credited as successful selections.

Noul: strong ranking did not guarantee safe acceptance

Noul was converted to a hard validity decision at the frozen threshold of 0.5. In answer-present sets, six candidates are valid and 66 invalid; an always-invalid classifier would already score 66/72 by ignoring every true solution. Accuracy alone would obscure the failure modes.

Candidate condition	True positives	Missed valid	False positives	True negatives	Brier score
Columns-only, answer present	2	4	34	32	0.2451
Explicit regions, answer present	6	0	5	61	0.0529
Explicit regions, no valid answer	0	0	4	68	0.0615

With explicit regions, the true candidate ranked highest on every answer-present board, but five invalid candidates also crossed the yes threshold. A higher score than the alternatives is not equivalent to an acceptable absolute probability threshold.

Brier score is the average squared difference between Noul's yes-probability and binary truth; lower is better. The always-invalid baseline has Brier 0.0833 on answer-present sets and 0 on all-invalid sets. Explicit regions improved that metric in the former condition; columns-only was much worse than the baseline. These small, clustered samples are insufficient to declare calibrated probabilities or choose a production threshold.

Score: ordinal precision was weaker than ranking

The three-level rubric separates multiple missing regions, one missing region and complete coverage. It intentionally tests whether Jev can discriminate degrees of constraint satisfaction, not merely choose a winner.

Candidate condition	Correct modal level	Mean absolute error, rubric units	Majority-level reference
Columns-only, answer present	12/72	0.940	37/72 by always choosing level 0
Explicit regions, answer present	32/72	0.542	37/72 by always choosing level 0
Explicit regions, no valid answer	30/72	0.605	41/72 by always choosing level 0

Even with explicit regions, exact ordinal classification was below a simple majority-level reference. This does not contradict 6/6 top-candidate ranking: the model can rank the fully valid option above the others while misjudging how far the invalid placements fall short.

When Score's probability on complete coverage was thresholded at 0.5, it accepted all six valid candidates but also four invalid ones in answer-present sets. It also accepted four invalid candidates across the all-invalid sets. The mean score and the probability of the top level should both be retained; they answer different questions.

Rejection when no solution is supplied

Set-level rejection rule	Correctly rejects all-invalid set
Choice selects none	6/6
Every candidate Noul is below 0.5	4/6
Every candidate Score probability of complete coverage is below 0.5	3/6

This is a useful distinct capability: an evaluation that always supplies one correct option cannot test rejection. It also shows why Choice is not automatically more challenging than Noul. In this experiment the single joint selection question was more reliable at set-level rejection than independently thresholding candidate judgments.

The [full per-candidate table](#) and [machine-readable values](#) expose all 216 candidate assessments, including Choice probability, Noul, Score, Score's complete-coverage probability and engine labels. Those files identify candidates and engine labels for all six new boards. Unlike the foundation board illustrations, this new-board section does not print solution coordinates.

Scientific interpretation of the new findings

The experiments separate at least five capabilities: reading board data, constructing a coordinated placement, selecting among prepared alternatives, evaluating absolute validity, and rating partial coverage. It would be misleading to call all five simply "solving Queens."

Candidate recognition is the demonstrated strength. Correct region-list choices recur across the original boards, larger option menus and six new boards. Preprocessing removes the coordinate-to-region indirection that repeatedly defeated the columns-only interface. Because instruction wording changes with the representation, the data support the combined interface rather than a formatting-only causal claim.

Planning remains unproven and weak in these tests. No native complete construction succeeded on the six new boards with either lossless text representation. Earlier local legal moves failed global completion. There is no evidence here that Jev can independently execute the search needed to discover the full solution.

Verification is not interchangeable with ranking. Noul and Score identify the best candidate when the correct one is available, yet assign acceptance-level values to some invalid candidates. The exact engine should reject those placements regardless of Jev's confidence.

Choice is not universally harder than a binary primitive. A balanced blind binary guess has a 50% reference probability, while a uniform guess among K alternatives has $1/K$. But our validity dataset is highly imbalanced, and Noul must support an absolute boundary, not just relative ordering. The candidate content, ambiguity, distractor quality, rejection option and evaluation rule determine difficulty more than the output type alone.

Score is not a calibrated intelligence meter. Its values depend on the authored levels, which are not physical distances between solutions. A one-unit difference is meaningful only within this rubric. Testing a richer ordinal question exposed weaknesses that winner selection had hidden; it did not establish a universal numerical measure of reasoning.

TypeSafe's [Jev 1.13 limitations](#) specifically warn about counting, numerical precision and multi-hop indirection, and advise keeping exact computations in code. They also caution that Noul and Choice probabilities need not obey cross-question identities and thresholds should not be transferred between primitives. Those vendor cautions are consistent with the observed interface sensitivity and scalar disagreements, but do not establish their internal cause or excuse an incorrect placement. The measured Queens failures still count as failures of the assigned tasks.

There are no formal population-level significance claims. The same board appears under several conditions; candidate lists overlap; prompts were developed in earlier stages; and all new boards share a generator family. Nominal random-pick references are helpful context but must not be multiplied across correlated trials to manufacture a tiny p-value.

What the comparison supports

The reasoning-model pilot demonstrated full written solution construction from a blank image within a single externally tool-free request. Jev demonstrated reliable selection only in the tested cases where code had made the remaining decision explicit and narrow. This is an observed task distinction, not a proof of a general architectural or intelligence hierarchy.

Capability	Evidence from this evaluation
Generate a globally valid solution from the blank image	All eleven returned reasoning-model answers were valid; one Grok result unavailable
Construct a globally valid placement from text	Initial constructions failed; matrix and prose each scored 0/6 on six new boards
Read a small textual grid	Jev got all nine sampled cell labels right
Recognize occupied-region distinctness among candidates	Initial 5x5 6/6; larger-board scale 24/24; six new boards 6/6 with explicit regions
Reject every invalid candidate at a fixed threshold	Noul and Score produced false positives despite correct top ranking
Plan a next move that permits eventual completion	Jev succeeded 0/6 in the tested one-given scenarios
Guarantee correctness	Deterministic validator remains authoritative

Speed and cost are conditional on the work assigned

Jev's subsecond calls are genuinely fast in the measured client path, but they are not measurements of end-to-end autonomous puzzle solving. Candidate generation, region lookups, exact-solution selection for test construction and preprocessing were outside those request timings. Sol's 13.70-second median included generating the whole placement from the supplied image. A ratio between those times would mix different tasks and omit Jev's helper work.

The repository solver is an explicit constraint engine, not a trained model. It was used to recover and validate reference solutions for the report, not run as an equivalently instrumented timed entrant. We therefore report **no solver-versus-model speed ratio** and do not call local compute free merely because it incurred no OpenRouter charge.

Practical implication

Keep the exact validator as the final gate. A Jev-assisted application could expose already computed candidates and clear local features, then use its choice as a proposal. For Queens specifically, once all constraints or region counts are available, ordinary code can finish the decision directly. This experiment has not shown an operational advantage for adding Jev to that exact-solving pipeline.

A fair follow-on would reuse the [existing comparative plan](#): equal text-input construction tests for all models; a separate identical-candidate recognition track; explicit tool-enabled and no-tool tracks; one-shot versus fixed-budget feedback attempts; held-out boards; and total pipeline time and cost. Held-out Jev boards and typed-primitives comparisons were subsequently run and are included here. Equal-input construction for all models, separate sandbox competitors and tool-enabled reasoning-model arms remain unrun. No new paid trials were launched for this editorial revision.

Next evaluation design

For the Queens application, retain the exact solver and checker as authoritative. Jev may be evaluated as a helper for bounded decisions, but do not claim it replaces the solver or independently validates every constraint when preprocessing already guarantees most rules.

The most useful next study is not additional retries on the same solved examples. It is a preregistered held-out evaluation with:

1. **Broader board sampling:** new boards from multiple generators or authorized source families, deduplicated geometrically and grouped by measurable solver difficulty, not size alone.
2. **Separate development and test sets:** select prompts and thresholds only on development boards; freeze them before the final set. Treat this report's boards as development evidence once inspected.
3. **Board-level analysis:** report one-shot success by board and format, with paired comparisons and uncertainty respecting board clusters. Do not treat all candidate judgments as independent trials.
4. **Absolute verification controls:** balanced valid/invalid examples, including isolated one-candidate requests, all-invalid sets, and different positive prevalences. This would test whether scalar judgments depend on surrounding candidate context.
5. **A frozen score rubric:** test single-dimension levels with known ground truth; retain both level distributions and expected scores. Evaluate ranking and exact level accuracy separately.
6. **Operational outcomes:** retain transport failures, unknown billing and retries; include preprocessing and checker time when judging a full system.

A separate multi-attempt track may allow checker feedback, backtracking or a candidate tournament exceeding 255 options. It should report success by attempt count, cumulative cost and time, and search work performed by code. It must not be relabeled as one-shot or merged with the current leaderboard. No such new paid study is launched by this recommendation.

Evidence definitions and reproduction

Measurement boundaries

- **Correct solution:** an in-range queen in every row, with unique columns, exactly one hit per region, and no touching pair. The unique-solution property is separately established by exhaustive enumeration.
- **Strict pass:** the original output also satisfies the exact JSON-only schema. Mathematical post-hoc extraction is a separate diagnostic, not a replacement primary score.
- **Elapsed API time:** monotonic client time through full response receipt, including routing/network/queuing. Not isolated model compute, time to first token, or end-to-end preparation time.
- **Confirmed USD:** returned `usage.cost`; missing charges are not zero. **Allowance:** conservative unreconciled reservation, not a billed amount.
- **One request:** one submitted model inference opportunity per original model/board. A pre-inference provider rejection is retained operationally but not counted as a wrong solution or a second model answer.

Limits that materially affect interpretation

The original comparison has only three synthetic scored boards, one trial per model/board, and one target for most early Jev follow-ups. Later Jev work adds six distinct boards from the same synthetic generator family; it does not add new original-model trials or establish broad distribution coverage. Size and difficulty are confounded. Clean images, region letters and single-cell regions simplify parts of perception/reasoning. The Gemini ID is a preview version. High effort and token limits are not equal compute, and latency conditions differed across providers and days. These observations do not establish population success rates, statistical superiority, a stable price ranking, or an architecture-level explanation.

The native Jev 7×7/9×9 probes occurred in the first deep dive because the initial output was interpretable. The later instruction not to advance beyond 5×5 was honored in the 39-call follow-up. The foundation report preserved both stages without claiming larger-board success. Later explicit user instructions authorized the scale and new-board studies; their successes concern prepared candidate recognition, not independent construction.

Inspectable evidence

- [Original protocol, results, request audit, input audit, and billing.](#)
- [Initial Jev assessment and audit.](#)
- [Jev input-variation assessment, frozen variants, follow-up design, and audit.](#)
- [Larger-board scale assessment and audit.](#)
- [New-board protocol, audit, candidate scalar tables, and offline tests.](#)
- [Foundation-era comparison evidence, board verification and image provenance,](#) and retained [illustrations](#). Exact submitted request bodies and raw outputs remain in their original evaluation folders. Raw Jev files remain local and Git-ignored.

Codex assembled the foundation report; GPT-5.6 Luna agents at Xhigh effort audited its evidence and prepared board illustrations. For this expanded edition, three new GPT-5.6 Luna Xhigh agents researched public sources, while root Codex integrated the retained evidence. They did not make new OpenRouter requests. Source results control over narrative summaries. The report intentionally reveals solution images and coordinates at the user's request; prior spoiler-free reports are preserved.

Seven offline tests and the saved primitive response audit passed. Mathematical judgments use the deterministic checker; LLM judgment is limited to interpretation of capability and limitations. The reporting skill informed that distinction, and the PDF skill required content and visual checks.

Private report; [publication terms need review](#). Local evidence links require project files; figures and narrative are embedded in the PDF.

Online source guide

Three GPT-5.6 Luna Xhigh agents reviewed these sources on September 19, 2026. Public pages explain interfaces; saved local evidence establishes Queens outcomes. Live catalogs do not replace experimental snapshots.

Model pages

- [OpenRouter: GPT-5.6 Sol](#).
- [OpenRouter: Claude Opus 4.8](#).
- [OpenRouter: Grok 4.6](#); saved serving route: amazon-bedrock/us-west-2.
- [OpenRouter: Gemini 3.1 Pro Preview](#).
- [OpenRouter: Jev 1.13](#) and [Jev Latest](#); distinguish the listing, moving alias and dated response build.

The original contestants used High effort. Xhigh describes the Luna researchers, not a change to contestant settings. Public catalogs do not validate this study's timing, billing or success measurements.

Jev definitions and limitations

- [Models and aliases](#) - text-only modality, version names and context constraints.
- [System One](#) and [direct API reference](#) - typed decisions and structured state; separate direct-provider and gateway routes.
- [Choice: fixed candidate selection](#) - relative option probabilities, the 255-option ceiling and rejection-option design.
- [Noul: probability of yes](#) - binary-question uncertainty rather than a middle quality level.
- [Score: ordered descriptive levels](#) - level distributions and their probability-weighted expected index.
- [Question primitives and batching](#) - shared state with separately evaluated questions, not proof of statistically independent puzzle trials.
- [Confidence](#) - distribution-based confidence, not a correctness certificate.
- [Jev 1.13 known limitations](#) - counting, indirection and cross-primitive inconsistency cautions relevant to interpreting these experiments.

Queens explanation and public play

Best authoritative description: [LinkedIn's Queens help](#). It gives the exact regional one-queen/no-touch rules used here, without confusing them with long-diagonal chess attacks.

Best primary play link: [LinkedIn Queens](#). A research agent verified Start game, board loading and cell toggling without signing in; it did not complete or submit a puzzle. [Guest-mode help](#) explains the account-free access and retention limits. Daily boards change, so this URL is not a frozen reproducibility artifact.

Practice alternative: [QueensGame.io](#) offers account-free, size-selectable boards. The agent observed its board and controls, not completion; ads were present. The user's deprovisioned Replit is not a working alternative.